SDM Institute of Technology, Ujire
Department of Information Science and Engineering
Major-Project Synopsis
6th Semester ISE-2026

Title	Design and Implementation of a Cloud-First Thin-Client Smart Keyboard Workstation
Abstract	This project proposes a cloud-first smart keyboard workstation for portable web development and automation. A Raspberry Pi operates as a thin client with a built-in mini projector for compact visual output. All development and computational tasks are handled through a cloud-hosted environment with AI-assisted automation. The system provides a low-cost, low-power, and scalable solution for student and field-based use.
Objectives	● To design a compact smart keyboard workstation using a Raspberry Pi and mini projector. ● To enable cloud-based development through a thin-client interface. ● To integrate AI-assisted agents for automated coding and task management. ● To develop a low-cost, low-power, and scalable portable development system.
Methodology	The system is built using a Raspberry Pi configured with a lightweight Linux operating system and connected to a compact mini projector for display. A browser-based thin-client interface is used to access a cloud-hosted development environment over the internet. AI-assisted automation agents are deployed in the cloud to perform coding, testing, and task execution. The entire setup is optimized to minimize power consumption, hardware cost, and local processing load.
Proposed outcome	The proposed system delivers a functional portable workstation for web development using minimal local hardware. It enables users to access powerful cloud-based development tools through a compact smart keyboard interface. AI-assisted automation improves productivity by handling repetitive coding and task management operations. The design significantly reduces power consumption and hardware cost compared to traditional laptops. The solution demonstrates a scalable and practical model for cloud-first development environments.
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